

Machine Learning

Programming and Music 2
<http://www.bjarni-gunnarsson.net>

Machine Learning

Machine Learning

Machine learning is a subfield of artificial intelligence (AI) that focuses on the development of algorithms and models that enable computer systems to learn and make predictions or decisions *without being explicitly programmed*.

Machine learning concerns building models that can learn from data. Instead of being explicitly programmed with specific rules or instructions, machine learning algorithms are trained on large amounts of data to learn patterns and extract meaningful information.

ML Types

There are different kinds of **Machine Learning types**.

1. Supervised Learning
2. Unsupervised Learning
3. Semi-Supervised Learning
4. Reinforcement Learning
5. Deep Learning
6. Combined approaches

Supervised Learning

In supervised learning, the algorithm learns from **labeled training data**, where each example is associated with a corresponding target or output value.

The goal is to learn a **mapping function** that can predict the output for new, unseen inputs.

Some popular algorithms in supervised learning include *linear regression*, *logistic regression*, *decision trees*, *random forests*, *support vector machines* (SVM), and neural networks.

Unsupervised Learning

Unsupervised learning involves learning from unlabeled data, where the algorithm tries to find patterns or structure in the data without any explicit target variable.

The goal is to discover **inherent relationships** or groupings in the data. Clustering and dimensionality reduction are common tasks in unsupervised learning.

Examples of unsupervised learning algorithms include k-means clustering, hierarchical clustering, principal component analysis (PCA), and generative adversarial networks (GANs).

Reinforcement Learning

Reinforcement learning involves an agent learning from interactions with an environment to maximize cumulative rewards. The agent takes **actions** in the **environment**, receives feedback in the form of rewards or penalties, and learns through trial and error.

The goal is to learn an **optimal policy** that maximizes the long-term expected reward.

Reinforcement learning has been successfully applied to various domains, including game playing, robotics, and autonomous systems.

Deep Learning

Deep learning is a subset of machine learning that focuses on training deep neural networks with multiple layers.

Deep neural networks are composed of interconnected layers of artificial neurons that can learn hierarchical representations from data. Deep learning has achieved remarkable success in areas such as computer vision, natural language processing, speech recognition, and many other domains.

Convolutional neural networks (CNNs) and **recurrent neural networks** (RNNs) are commonly used in deep learning.

Neural Networks

Neural Network

A ***neural network*** is a computational model inspired by the structure and functioning of the biological brain. It is a network of interconnected artificial neurons, also called nodes or units, that work together to process and learn from input data to produce output or make predictions.

The basic building block of a neural network is the ***artificial neuron***, often referred to as a ***perceptron***. Each neuron takes one or more input values, applies weights to those inputs, and passes the weighted sum through an activation function.

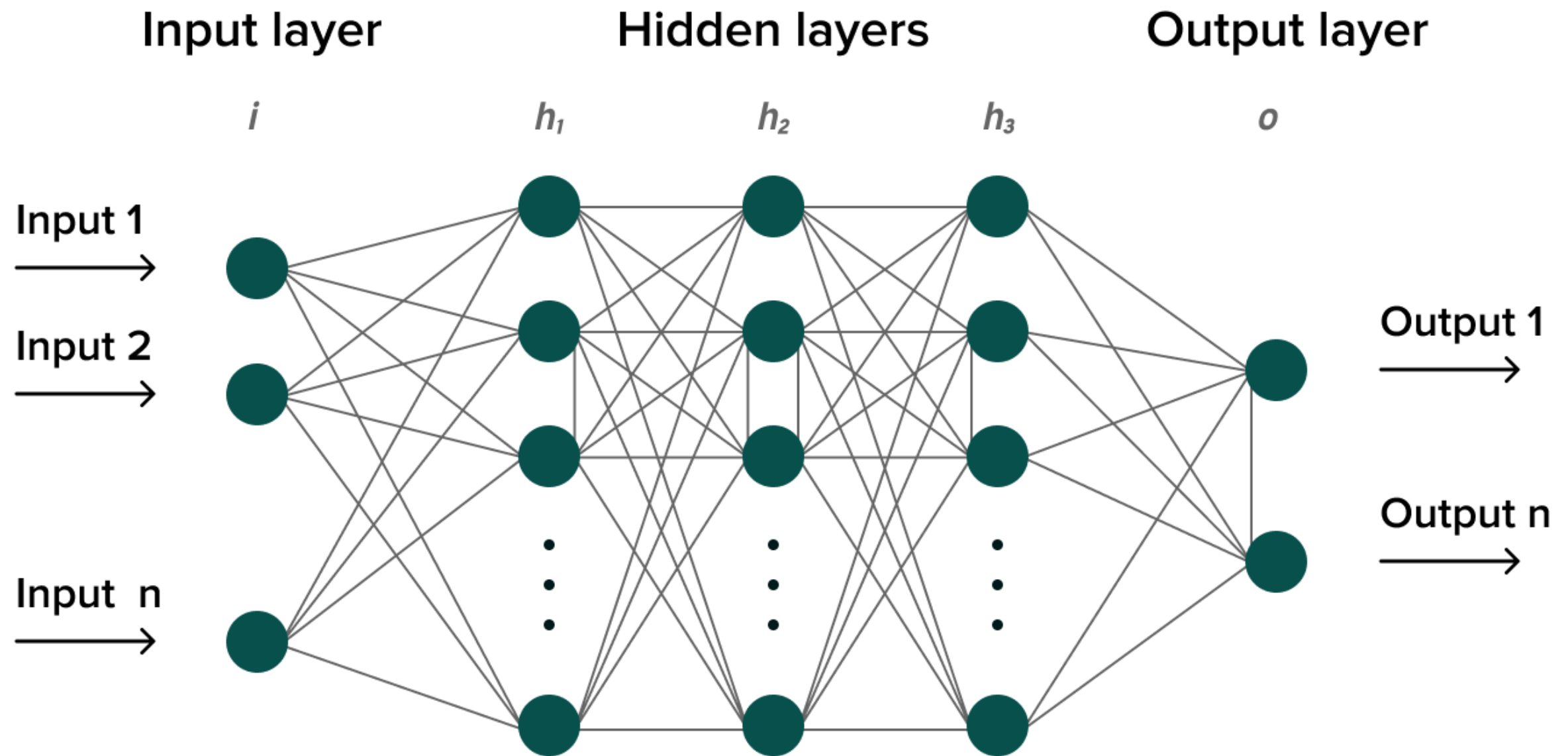
Neural Network

Input Layer: The input layer receives the initial input data and passes it to the subsequent layers. Each input node corresponds to a feature or attribute of the data.

Hidden Layer(s): Hidden layers are intermediate layers between the input and output layers. They perform computations and learn representations of the input data.

Output Layer: The output layer produces the final output or predictions of the neural network.

Neural Network



Neurons

Step by step:

1. **Inputs:** several input values (like numbers for data features).
2. **Weights:** Each input is multiplied by a weight (importance).
3. **Summation:** It adds up all the weighted inputs.
4. **Activation:** It passes the sum through a simple function (e.g., if the sum is above a threshold, output 1; otherwise, 0).

Mathematically:

output = 1 if $(w_1 \times x_1 + w_2 \times x_2 + \dots + w_n \times x_n + \text{bias}) > \text{threshold}$, else 0

The perceptron “learns” by adjusting the weights when it makes mistakes for better decisions next time.

Classification

Classification

Classification using a neural network is a machine learning task where the goal is to assign input data into predefined categories or classes.

In *FluComa*, **MLPClassifier** is a neural network that can be used to perform classification, an algorithm that is trained to predict what category, or class, an input belongs to. “MLP” stands for multi-layer perceptron which is a type of neural network.

By providing input data as a ***DataSet*** and output labels as a ***LabelSet***, the neural network is trained using supervised learning so that it can predict the most probable label for a given data point input.

MFCC

MFCC is used to represent the timbre or spectral shape of an audio signal in a compact form.

It uses **FFT**, **windowing** and the **Mel Scale**: a perceptual scale of pitch that reflects how humans hear frequency (pitching is linear at low frequencies and logarithmic at high ones). A **Mel filterbank** is a series of overlapping triangular filters applied to the frequency spectrum to mimic this perceptual sensitivity.

Produces a small number (typically 12–13) of **coefficients** per frame that effectively describe the spectral envelope.

Exercises

- Exercises -

1. *Extend the code only example to use your own Sounds*
2. *Extend the GUI Example to to use your own Sounds*
3. *Adapt either examples to a mapping that are not only sounds synthesis ones.*

